

NGUYEN ANH MINH

📍 Hanoi, Vietnam

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ABOUT ME

I am seeking PhD opportunities in Natural Language Processing and Large Language Models, with a primary focus on the reasoning capability of LLMs and Retrieval-augmented generation (RAG). My recent work investigates how LLMs can perform retrieval and multi-hop reasoning directly in hidden representation space, bypassing explicit text-based intermediate steps. I am also actively working on GraphRAG and LLM evaluation for specialized domains, and maintain broader interests in Information Retrieval and Recommendation Systems grounded in my prior research on LLM-enhanced and graph-based recommendation.

EDUCATION

Hanoi University of Science and Technology (HUST)

Aug 2019 - October 2023

Bachelor's Degree, Applied Mathematics and Informatics.

- Valedictorian of Applied Mathematics and Informatics Talented Program.

GPA: 3.78/4.00.

HONORS AND AWARDS

First Prize, Viettel AI Challenge, 2024.

- Achieved 1st place among **300 AI engineers in Viettel** by developing a top-performing product *recommendation system* for an e-commerce platform.

First Prize, Viettel Digital Talent, 2023.

- Secured 1st place among **8,000 students** with the *Cross-Domain recommendation system* project.

First Prize, Econometrics and Applications Olympiad, 2022.

- Awarded by the Institute of Mathematics, VAST, Hanoi, for securing 1st place among **400 teams**. Developed a *Bonus-Malus System model* for insurance pricing using Markov Chains.

PUBLICATIONS

1. **Nguyen, Minh-Anh**, and Dung D. Le. *Learning Effective Language Representations for Sequential Recommendation via Joint Embedding Predictive Architecture*. Conference on Language Modeling (**COLM**), 2025, [Link](#).
2. **Nguyen, Minh-Anh**, Bao Nguyen, Ha Lan N.T., Tuan Anh Hoang, Duc-Trong Le, and Dung D. Le. *Enhancing Graph-based Recommendations with Majority-Voting LLM-Rerank Augmentation*. **Machine Learning Journal** (Springer Q1), 2026, [Link](#).
3. **Nguyen, Minh-Anh**, Minh-Duc Nguyen, Nguyen Thi Ha Lan, Kieu Hai Dang, Nguyen Tien Dong, and Dung D. Le. *SurveyG: A Multi-Agent LLM Framework with Hierarchical Citation Graph for Automated Survey Generation*. **AI4X**, 2026; **ICML AI4Science Workshop**, 2026. [Link](#).
4. **Nguyen, Minh-Anh**, Nguyen Tien Dong, Thanh Dat Hoang, Nguyen Tuan Ngoc, Dao Xuan Quang Minh, Phan Phi Hai, Nguyen Thi Ngoc Anh, and Binh Vu. *VLegal-Bench: Cognitively Grounded Benchmark for Vietnamese Legal Reasoning of Large Language Models*. **ICML AI4Law Workshop - Oral**, 2026. (Supervised by Dr. Nguyen Thi Ngoc Anh (HUST) and Prof. Binh Vu (SRH Heidelberg, Germany).) [Link](#).
5. **Nguyen, Minh-Anh**, Ha Lan N.T., and Dung D. Le. *LAnR: Latent Abstraction Retrieval-Augmented Generation for Multi-hop Reasoning*. **ICML EMM-QA Workshop**, 2026, [Link](#).

WORK EXPERIENCE

CMC OpenAI

August 2025 - Present

AI Research Expert

Research on LLM Evaluation, Graph-based RAG, and Legal NLP

- Leading a team of 4 researchers on *VLegal-Bench*, the first public cognitively grounded benchmark for Vietnamese legal reasoning in LLMs, in academic collaboration with Dr. Nguyen Thi Ngoc Anh (HUST) and Prof. Binh Vu (SRH Heidelberg, Germany). Accepted ORAL at **ICML AI4Law 2026**.
- Conducting applied research on Graph-based Retrieval-Augmented Generation (GraphRAG), evaluating and adapting recent state-of-the-art methods for large-scale Vietnamese legal knowledge bases.

VinUni

February 2025 - Present (part-time from August 2025)

Research Assistant, advised by Prof. Dung D. Le

Research on Large Language Models, Latent Reasoning, and Recommendation Systems

- **Latent Reasoning in RAG (LAnR):** Proposed a novel framework in which LLMs perform retrieval directly in latent representation space via special [PRED] tokens, eliminating explicit query generation. Designed an entropy-guided MLP control head for adaptive multi-hop retrieval termination, targeting improved accuracy and reduced inference latency over state-of-the-art iterative RAG baselines (Search-R1, Self-RAG, IRCOT).
- **LLM-Enhanced Recommendation Systems:** Led two first-author projects integrating LLMs with recommendation: (i) a Joint Embedding Predictive Architecture for sequential recommendation (COLM 2025), and (ii) a majority-voting LLM-rerank augmentation for graph-based recommendation (Machine Learning Journal, Q1).
- **Multi-Agent LLM Systems:** Developed SurveyG, a multi-agent LLM framework with hierarchical citation graphs for automated scientific survey generation (under review at KDD 2026).

Viettel Telecom

October 2022 - February 2025

Data Scientist

Recommendation and Search Systems for TV360 - Vietnam's National Digital Television Platform (200K daily active users)

- Designed and deployed a **Graph-based Recommendation System** for movies and TV programs, serving approximately **100K daily active users** (50% of the TV360 user base).
- Developed the **core search module for TV360**, including an **ElasticSearch** index for content retrieval and an LLM-based user intent recognition component, served at scale with **Ray Serve + vLLM**; deployed to the same **100K DAU** user segment.
- Built an **ETL pipeline** using **Hive SQL** to aggregate customer and product data feeding both the recommendation system and executive reporting dashboard.
- Implemented production recommendation rules for videos and channels using **PySpark** on a distributed big-data stack.

REFERENCE

Dr. Le Duy Dung

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Dr. Nguyen Thi Ngoc Anh

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